

Solution Requirements

2. Functional Requirements (FR)

The system must deliver the following capabilities, structured by user access level (Role-Based Access Control):

1. User & Authentication Services

- **FR-1.1:** Standard users must be able to register an account with a unique email, name, and hashed password.
- **FR-1.2:** Users must be able to log in securely to retrieve a signed JSON Web Token (JWT) session.
- **FR-1.3:** The system must restrict administrative dashboard routes to users holding the `admin` or `seller` roles.
- **FR-1.4:** Standard users must be able to manage an address book containing multiple shipping locations, selecting a default address for checkouts.

2. Product Catalog & Discovery

- **FR-2.1:** The database must list products displaying name, description, price, stock inventory, categories, and images.
- **FR-2.2:** The frontend must provide full-text search with live autocomplete suggestions.
- **FR-2.3:** The interface must support dynamic filters based on Category, Price Range, Minimum Ratings, and Stock Availability.
- **FR-2.4:** The frontend must support sorting configurations: Price (Low to High, High to Low), Ratings, and Release Date (Newest).

3. Shopping Cart & Wishlist Lifecycle

- **FR-3.1:** The system must offer a persistent shopping cart linked to the user context.
- **FR-3.2:** Users must be able to modify item quantities in the cart, subject to real-time stock limits.
- **FR-3.3:** The system must maintain a saved Wishlist, permitting users to move saved favorites to the cart in a single click.

4. Checkout, Coupons & Payments

- **FR-4.1:** The system must guide the customer through a multi-step checkout wizard (Shipping Address -> Shipping Speed -> Coupon Apply -> Mock Payment -> Order Confirmation).
- **FR-4.2:** The coupon engine must validate active coupon codes, applying percentage or fixed discounts and recalculating itemized taxes and shipping fees.
- **FR-4.3:** The checkout must communicate with a mock payment intent endpoint to securely complete transactions.

5. Administrative Operations

- **FR-5.1:** Admins and sellers must hold complete CRUD (Create, Read, Update, Delete) privileges over products.
- **FR-5.2:** Sellers and admins must be able to upload product images, storing them on Cloudinary.
- **FR-5.3:** Admins must be able to modify order fulfillment milestones (e.g., Placed -> Confirmed -> Packed -> Shipped -> Out for Delivery -> Delivered).
- **FR-5.4:** Admins must be able to block/unblock customer accounts, instantly revoking active sessions.
- **FR-5.5:** Admins must be able to manage coupon campaigns (create, toggle active state, delete).

3. Non-Functional Requirements (NFR)

To ensure the shopEZ platform is secure, responsive, and robust, it must comply with these metrics:

1. Security & Protection

- **NFR-1.1:** Passwords must be hashed using the Bcrypt algorithm before database commits.
- **NFR-1.2:** Session JSON Web Tokens must be transmitted via HTTP-Only, Secure cookies to prevent XSS-based token extraction.
- **NFR-1.3:** Enforce strict API middleware validation to deny unauthenticated access to protected routes.

2. Performance & Optimization

- **NFR-2.1:** The application UI must maintain high responsiveness, utilizing React's virtual DOM to update components instantly without full-page reloads.
- **NFR-2.2:** Product images must be stored and distributed via Cloudinary's global Content Delivery Network (CDN) to ensure quick image loading.
- **NFR-2.3:** API responses for the catalog must complete within 200ms under standard loads.

3. Reliability & Integrity

- **NFR-3.1:** Database commits during checkout must ensure atomic stock deductions to prevent oversells.

- **NFR-3.2:** If the primary MongoDB cluster connection is unavailable, the backend seeder must automatically instantiate a local In-Memory MongoDB fallback to guarantee development and staging continuity.

4. Usability & Accessibility

- **NFR-4.1:** The user interface must utilize a fully responsive layout that fits mobile, tablet, and desktop viewports.
- **NFR-4.2:** Design styling must utilize Outfit or Inter fonts with clear typography hierarchies.